Science & Sports 39 (2024) 516—519

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SHORT COMMUNICATION

Effect of load reductions on performance of a strength training session with different exercise orders

Effet des réductions de charge sur la performance d'une séance de musculation avec différents ordres d'exercices

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Received 13 August 2021; accepted 9 May 2023
Available online 18 May 2024

KEYWORDS
Resistance training;
Training volume;
Volume load;
Performance

Summary
Introduction. — When strength training (ST) sets are performed to muscular failure over successive sets, a reduction in performance is expected, which would compromise training volume (TV). One method to sustain repetitions and limit the volume impairment is to acutely reduce the load over successive sets. However, previous studies did not investigate this load reduction in different exercise orders. The aim of the present study was to investigate the effect of progressive 10% load reductions on TV with different exercise orders.
Summary of facts and results. — Eleven healthy men performed four protocols: multi- to single-joint (MS) protocol and single- to multi-joint (SM) protocol were performed with a constant 10-RM load or with a 10% load reduction over successive sets (MS10 and SM10), respectively. The results showed that the two load reduction protocols had significantly greater TV versus the two protocols without load reduction, independent of the order ($P < 0.05$). Significantly greater total volume was achieved on the last exercise of a sequence with load reduction versus without load reduction ($P > 0.05$).
Conclusion. — Training volume can be maintained more effectively when a load reduction of 10% is progressively applied over successive sets.
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https://doi.org/10.1016/j.scispo.2023.05.002
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MOTS CLÉS
Entraînement en résistance ;
Volume d'entraînement ;
Charge volumique ;
Performance

Résumé
Introduction. — Lorsque des séries de musculation (ST) sont effectuées jusqu'à l'échec musculaire sur des séries successives, une réduction des performances est attendue, ce qui compromettrait le volume d'entraînement (TV). Une méthode pour maintenir les répétitions et limiter la perte de volume consiste à réduire fortement la charge sur des séries successives. Cependant, les études précédentes n'ont pas étudié cette réduction de charge dans différents ordres d'exercice. Le but de la présente étude était d'étudier l'effet de réductions de charge progressives de 10 % à la télévision avec différents ordres d'exercice.
Résumé des faits et des résultats. — Onze hommes en bonne santé ont réalisé quatre protocoles : le protocole multi-articulation à simple (MS) et le protocole mono-articulation (SM) ont été réalisés avec une charge constante de 10-RM ou avec une réduction de charge de 10 % sur des ensembles successifs (MS10 et SM10), respectivement. Les résultats ont montré que les deux protocoles de réduction de charge avaient un TV significativement plus élevé par rapport aux deux protocoles sans réduction de charge, indépendamment de l'ordre ($p < 0{,}05$). Un volume total significativement plus important a été atteint lors du dernier exercice d'une séquence avec réduction de charge par rapport à sans réduction de charge ($p > 0{,}05$).
Conclusion. — Le volume d'entraînement peut être maintenu plus efficacement lorsqu'une réduction de charge de 10 % est appliquée progressivement sur des séries successives.
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1. Introduction

Volume is a key variable in strength training (ST) that has a dose-response relationship with strength gains and muscular hypertrophy [1]. Strength training variables can directly influence TV such as the load, exercise order, and the rest interval between sets. Acute ST performance with different exercise orders showed an impairment in TV for those exercises performed at the end of a session [2]. When ST sets are performed to muscular failure over successive sets with an absolute load, a reduction in repetition performance is expected due to muscle fatigue, which would compromise TV [3]. One method to sustain repetitions and limit the volume impairment is to acutely reduce the load over successive sets. Willardson et al. [4] investigated 10% load reductions for trained women in performing three repetition maximum sets. The authors observed that this percentage reduction was sufficient to maintain the repetitions per set with a one-minute rest interval between sets. However, the previous study investigated only multi-joint exercises, which may not reflect the practical application of ST that often involves single-joint exercises as well. In this sense, the influence of exercise order has not been investigated with regard to acute load reductions in the session TV. Schoenfeld et al. [5] showed in a previous study that sessions with greater TV, as observed in those with longer rest interval between sets will lead to superior muscle adaptations such as hypertrophy, strength, and endurance. Thus, it is important to understand the interaction between exercise order and acute load reductions in the ability to maintain repetition performance over successive sets, in order to improve the session TV. Therefore, the purpose of the present study was to determine the effect of acute load reductions over successive sets for a ST session performed in opposite orders.

2. Methods

Eleven healthy men (24.45 ± 5.21 years, 74.4 ± 13.22 kg, 1.74 ± 0.05 m, 24.5 ± 4.27 BMI) participated in this study. The first two visits consisted of ten repetition maximum (10-RM) test and retest trials [2] in the following exercises: bench press (BP), shoulder press (SP), and triceps extension (TE). During a test, a rest interval between trials ranged between three to five exercises and between exercises was 10 minutes. Intraclass correlation coefficients between test and retest were 0.85 for the BP, 0.91 for the SP, and 0.95 for the TE.

The last four visits involved performance of four different workout formats in an alternating entrance design: the multi- to single-joint (MS) protocol (i.e., bench press, shoulder press, and triceps press) and single- to multi-joint (SM) protocol (triceps press, shoulder press, bench press) were performed with a constant 10-RM load or with a 10% load reduction over successive sets (MS10 and SM10), respectively. Three sets with a one-minute rest interval were performed for all exercises in each protocol. The number of repetitions performed in each set of each exercise and the TV (sum of repetitions × load).

A one-way ANOVA was applied with Bonferroni post-hoc to verify if there were differences between groups in relation to TV. For all inferential analyses, a $P < 0.05$ was adopted. Cohen's $d_{average}$ effect size (ES) and 95% CI was calculated with Hedges' $g_{average}$ correction.

3. Results

For the BP exercise, one-way ANOVA revealed significant differences between treatment groups ($F = 38{,}630$; $P < 0.001$). Bonferroni pairwise comparison showed significantly less TV for SM versus the MS ($P < 0.001$; $d = 2.09$), MS10 ($P < 0.001$; $d = 3.30$), and SM10 ($P < 0.001$; $d = 1.71$) protocols. The MS10 protocol resulted in significantly greater TV versus the MS ($P < 0.001$; $d = 1.95$) and SM10 ($P < 0.001$; $d = 1.45$) protocols.

For the SP exercise, one-way ANOVA revealed significant differences between treatment groups ($F = 13{,}660$; $P < 0.001$). Bonferroni pairwise comparison showed significantly greater TV for SM10 versus the MS ($P < 0.001$; $d = 1.68$), MS10 ($P = 0.01$; $d = 0.80$), and SM ($P = 0.002$; $d = 0.69$)

Table 1 Total volume, in kilograms, in different protocols and exercises.

	Bench press	Shoulder press	Triceps extension	Session
MS	1254 ± 149[c]	887 ± 192	902 ± 181	3043 ± 345
MS10	1678 ± 248[a,c,d]	1113 ± 272[a]	1442 ± 273[a]	4233 ± 682[a,c]
SM	832 ± 221	1062 ± 269	1351 ± 287[a]	3245 ± 625
SM10	1277 ± 256[c]	1386 ± 350[a,b,c]	1698 ± 291[a,b,c]	4361 ± 779[a,c]

MS: multi- to single-joint; SM: single- to multi-joint; MS10: multi- to single-joint with load reduction; SM: single- to multi-joint with load reduction.
[a] Significantly greater volume versus the MS protocol.
[b] Significantly greater volume versus the MS10 protocol.
[c] Significantly greater volume versus the SM protocol.
[d] Significantly greater volume versus the SM10 protocol.

Table 2 Effect size, mean difference, 95% confidence interval and magnitude of protocols comparison between exercises and session.

	Bench press		Shoulder press		Triceps extension		Session	
	ES (95% CI)	Mean difference (95% CI)	ES (95% CI)	Mean difference (95% CI)	ES (95% CI)	Mean difference (95% CI)	ES (95% CI)	Mean difference (95% CI)
MS vs. MS10	1.95 (0.97; 2.63) Large	−423.5 (−614.6; −232.3)	0.89 (0.23; 1.44) Moderate	−226.4 (−445.9; −6.8)	2.17 (1.12; 2.89) Large	−540 (−756.4; −323.6)	2.12 (1.08; 2.82) Large	−1189.8 (−1693.6; −686)
MS vs. SM	2.09 (1.06; 2.78) Large	422.2 (246.2; 598.2)	0.69 (0.08; 1.24) Small	−175.3 (−393.3; 42.7)	1.75 (0.84; 2.40) Large	−448.9 (−672.8; −225)	0.36 (−0.20; 0.91) Small	−202 (−672.7; −268.7)
MS vs. SM10	0.10 (−0.44; 0.67) Trivial	−22.5 (−217.7; 172.6)	1.68 (0.80; 2.32) Large	−499.1 (−762.6; −235.6)	3.08 (1.69; 3.97) Large	−795.9 (−1022.2; −569.6)	2.14 (1.10; 2.85) Large	−1317.5 (−1879.2; −755.9)
MS10 vs. SM	3.30 (1.82; 4.23) Large	845.6 (626.2; 1065)	0.17 (−0.37; 0.73) Trivial	−51.1 (−201.1; −303.3)	0.30 (−0.26; 0.85) Small	91.1 (−170.3; 352.5)	1.38 (0.59; 1.98) Moderate	987.8 (377.9; 1597.7)
MS10 vs. SM10	1.45 (0.64; 2.06) Moderate	400.9 (165.9; 635.9)	0.80 (0.17; 1.35) Small	−272.7 (−565.1; 19.7)	0.83 (0.19; 1.38) Moderate	−255.9 (−519.4; 7.6)	0.16 (−0.38; 0.72) Trivial	−127.7 (−810.3; 554.8)
SM vs. SM10	1.71 (0.81; 2.34) Large	−444.7 (−667.6; −221.9)	0.96 (0.28; 1.51) Moderate	−323.8 (−615.1; −32.6)	1.10 (0.39; 1.66) Moderate	−347 (−616.7; −77.3)	1.45 (0.64; 2.06) Moderate	−1115.5 (−1774; −457.1)

MS: multi- to single-joint; SM: single- to multi-joint; MS10: multi- to single-joint with load reduction; SM: single- to multi-joint with load reduction; ES: effect size; CI: confidence interval.

protocols. For the MS10 protocol, significantly greater TV was observed versus the MS ($P = 0.046$; $d = 0.89$) protocol.

For the TE exercise, one-way ANOVA showed significant differences between treatment groups ($F = 37{,}468$; $P < 0.001$). Bonferroni pairwise comparison revealed significantly less TV for MS versus the MS10 ($P < 0.001$; $d = 2.17$), SM ($P < 0.001$; $d = 1.75$), and SM10 ($P < 0.001$; $d = 3.08$) protocols. For the SM10 protocol, significantly greater TV was observed versus the MS10 ($P = 0.014$; $d = 0.83$) and SM ($P < 0.001$; $d = 1.10$) protocols.

When considering the TV of the entire session, the protocols with reduced load, MS10 and SM10, resulted in significantly greater performance versus the MS and SM ($F = 40{,}760$; $P < 0.001$) protocols (Tables 1 and 2).

4. Discussion

The main finding of the present study was the load reduction is a good strategy to improve TV in a session compared to traditional conditions, which generally use the same load throughout sets. The MS10 and SM10 protocols resulted in significantly greater TV versus the MS and SM protocols, explained by the greater number of repetitions performed

in the load reduction protocols. The protocols with greater TV can lead to superior adaptations, as showed by Longo et al. [6] that compared investigated the effects of different rest intervals (1-min vs. 3-min), with volume-equated or not in recreationally-trained men. The results showed that ST performed with 3-min rest interval shows greater TV compared to 1-min. Otherwise, the results revealed that the ST performed with 1-min but with the same TV of 3-min group shows similar muscle adaptations of hypertrophy and strength. Thus, it is important to perform ST sessions with variables manipulation that does not compromise the TV. Practically speaking, if the prescription calls for three sets of a given exercise, reducing the load 10% after the first and second sets could be used as a method to obtain higher TV without increasing the number of sets. Another important result of this study was in relation to TE exercise, for which the TV of the SM session was similar to the TV of the MS10 session. A similar effect was noted for the BP exercise, in which the TV of MS session was similar to the TV of the SM10 session. This could be used as a volume-up strategy for the fatigue-induced impairment of the triceps extension and bench press when they are placed as the last exercise of a training session.

Although the load reduction seems to be a good strategy to achieve a higher TV without greatly increasing session time, and benefiting endocrine responses, there is a lack of studies in literature. The percentage of reduction of this study was chosen from a previous study that used exercises for upper limbs such as the BP [7]. Nevertheless, the 10% was not enough to maintain the 10-repetition target for each set, and the last set being impaired in all exercises of all protocols. It is important to remember that the current study was performed with men, while the Willardson et al. [7] investigation was performed with women, who fatigue less, and require shorter recovery times than men [8].

Therefore, a load reduction protocol could be used as a method or strategy to increase the TV of a ST session, especially in the last exercises. In conclusion, recreationally trained subjects may use a load reduction of 10% following the first and second sets when training with three sets of 10 repetitions and a one-minute rest interval between sets.

Disclosure of interest

The authors declare that they have no competing interest.

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